



# INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

## Mid-Autumn Semester Examination 2022-23

Date of Examination: \_\_\_\_\_ Session: (FN/AN) \_\_\_\_\_ Duration: 2 hrs Full Marks: 30  
 Subject No.: PH 31207 Subject: CLASSICAL MECHANICS  
 Department/Center/School: \_\_\_\_\_

### 1. Spherical Pendulum

- Consider a spherical pendulum in three dimensions. Write down the Lagrangian in Cartesian coordinates given that the force due to gravity is  $\vec{F} = -mg\hat{z}$
- Assuming that the length of the pendulum remains fixed at  $r_0$  obtain the Lagrangian in the spherical polar coordinates using the following transformation laws,

$$x = r_0 \cos \phi \sin \theta; y = r_0 \sin \phi \sin \theta, z = -r_0 \cos \theta$$

- Identify the cyclic coordinate and derive the conserved conjugate momentum  $J$
- Argue the conservation of  $J$  from the Noether's theorem
- Show that the equation of motion of the  $\theta$  coordinate can be written in the following form:

$$\frac{d^2\theta}{dt^2} = -\frac{\partial V_{eff}(\theta, J)}{\partial \theta}$$

Obtain the form of  $V_{eff}(\theta, J)$

[3+2+2+2+5]

### 2. Conservative system in 1d

- Write down the action ( $S$ ) in terms of the Lagrangian ( $L$ ).
- What is the condition on the Lagrangian ( $L$ ) that extremizes the action.
- Consider the following 1d conservative Lagrangian

$$L(\dot{x}, x) = \frac{1}{12}m^2\dot{x}^4 + m\dot{x}^2V(x) - V(x)^2$$

- Write down the equation that will extremize the action for this Lagrangian
- Derive the momentum ( $P_x$ ) conjugate to the coordinate variable ( $x$ )
- Derive the corresponding Hamiltonian from this Lagrangian using the Legendre transformation and express it in terms of the usual 1d momentum  $p_0 = m\dot{x}$

[1+1+(5+2+3)]

- For a simple harmonic oscillator, the Lagrangian is given by  $L = \frac{\dot{q}^2}{2} - \frac{q^2}{2}$ . If  $A_{\pm}(q, p) = \frac{1}{\sqrt{2}}(p \pm iq)$  and  $H(q, p)$  be the Hamiltonian of the system then evaluate the Poisson bracket  $\{A_{\pm}(q, p), H(q, p)\}$ .

[4]